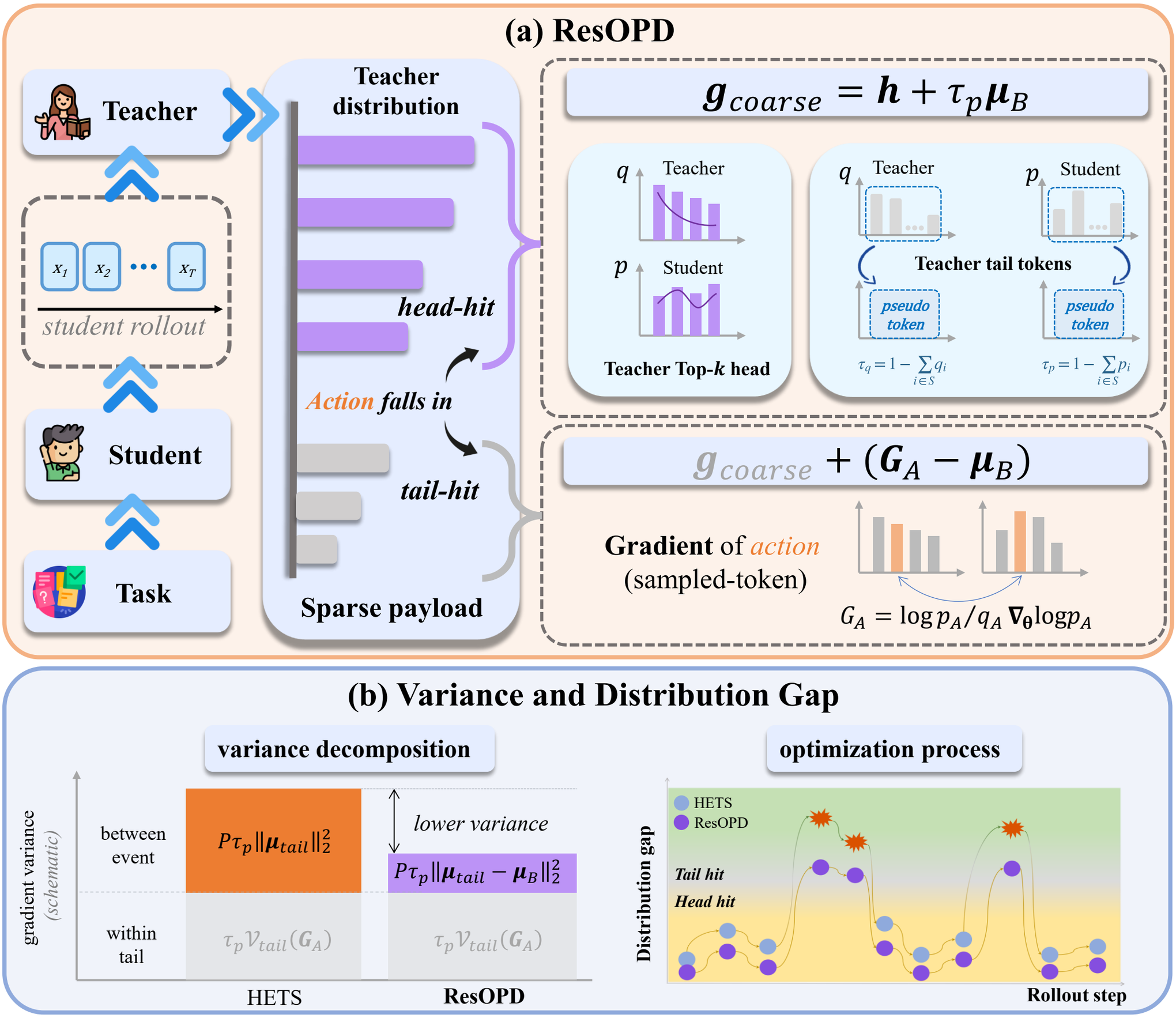

(a) ResOPD
Teacher
Student
Task
x_1 x_2 ••• x_T
student rollout
Teacher distribution
head-hit
Action falls in
tail-hit
Sparse payload
$g_{coarse} = h + \tau_p \mu_B$
q Teacher
p Student
Teacher Top-k head
Teacher tail tokens
pseudo token
pseudo token
$\tau_q = 1 - \sum_{i \in S} q_i$
$\tau_p = 1 - \sum_{i \in S} p_i$
$g_{coarse} + (G_A - \mu_B)$
Gradient of action (sampled-token)
$G_A = \log p_A / q_A \nabla_\theta \log p_A$
(b) Variance and Distribution Gap
variance decomposition
gradient variance (schematic)
between event
within tail
$P\tau_p \|\mu_{tail}\|_2^2$
$P\tau_p \|\mu_{tail} - \mu_B\|_2^2$
$\tau_p \mathcal{V}_{tail}(G_A)$
$\tau_p \mathcal{V}_{tail}(G_A)$
lower variance
HETS
ResOPD
optimization process
Distribution gap
HETS
ResOPD
Tail hit
Head hit
Rollout step